2016.0 RANGE ROVER (LG), 419-01

ANTI-THEFT - ACTIVE

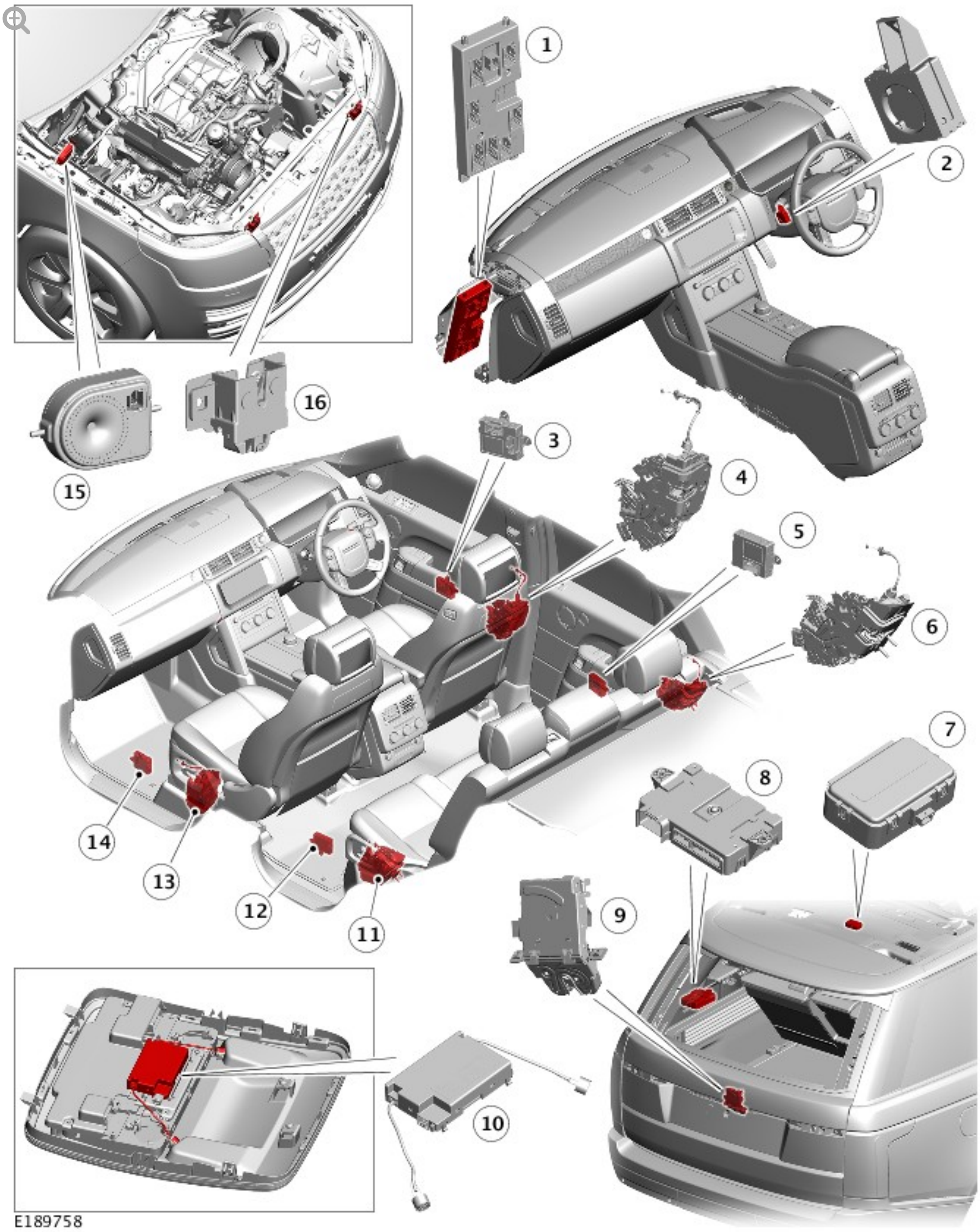
DESCRIPTION AND OPERATION

COMPONENT LOCATION

COMPONENT LOCATION

NOTE:

Right Hand Drive (RHD) installation shown, Left Hand Drive (LHD) installation similar



ITEM	DESCRIPTION
1	Body Control Module/Gateway Module (BCM/GWM)
2	Immobilizer Antenna Unit (IAU)
3	Driver Door Module (DDM)
4	Door ajar switch - Driver
5	Rear Door Module (RDM)
6	Door ajar switch - Rear

7	Radio Frequency (RF) receiver
8	Remote Function Actuator (RFA)
9	Tailgate ajar switch
10	Interior motion sensor (in overhead console)
11	Door ajar switch - Rear
12	Rear Door Module (RDM)
13	Door ajar switch - Passenger
14	Passenger Door Module (PDM)
15	Battery Back-up Sounder (BBUS)
16	Hood switch

OVERVIEW

The anti-theft - active system monitors all doors, hood and tailgate for unauthorized opening. In some markets the anti-theft system also incorporates monitoring of the vehicle interior and vehicle tilt sensing.

The anti-theft - active system is controlled by the following body system control modules:

- Body Control Module/Gateway Module (BCM/GWM) assembly
- Driver Door Module (DDM)
- Passenger Door Module (PDM)
- Rear Door Module (RDM) (2 off)
- Instrument Cluster (IC)
- Remote Function Actuator (RFA).

The BCM/GWM assembly is the main controller in the system. The BCM/GWM assembly controls the following security functions, in addition to other vehicle functions:

- Single locking, double locking (market dependant) and unlocking (both central unlock and single point entry)
- Monitoring of hood switch, door ajar switch and the tailgate ajar switch of the Central Door Locking (CDL) system
- Interior motion sensor
- Smart Key transponder reading
- Security Sounder - Battery Back-Up Sounder (BBUS) or Passive sounder
- Passive arming and disarming
- Panic alarm function
- Interior lighting.

There are two levels of vehicle anti-theft available:

- Perimeter sensing mode - Monitors all opening panels.
- Perimeter with alarm sensing mode - Monitors the vehicle interior for intrusion and it also incorporates a tilt sensor to monitor if the vehicle is being moved.

NOTE:

Volumetric mode and tilt sensing are not available in certain markets

Perimeter Sensing Mode

All apertures (doors, tailgate, hood) are monitored for ajar status (doors ajar and latch status change) and a visual (via directional indicators) and audible alarm is emitted. This is the via security sounder and vehicle horns (market dependent) in the event of unauthorized access.

The alarm is also triggered in the event of an unauthorized start request.

Smart Key

The Smartkey provides the following functionality:

- Unlock (central unlock or single point entry)
- Single lock and double lock (Market dependant)
- Tailgate release
- Approach lighting
- Panic alarm.

The lock and unlock switches also control a 'lazy' lock and unlock feature which will automatically close or open the windows with an extended press of the applicable switch. This feature is only available in certain markets and is controlled in conjunction with the door modules.

WARNING:

Never double lock the vehicle with any person or animal inside.

NOTE:

Double locking is only available in certain markets.

The smart key contains an emergency access key. This can be used in the event of failure of the smart key or the vehicle battery to unlock the vehicle. The driver door handle contains a concealed mechanical key barrel which can be used with the emergency key to access the vehicle. This will not disable the perimeter or alarm sensors systems and the visual and audible alarm will be activated when the door is unlocked/opened. To cancel the alarm, the smart key must be held next to the Immobilizer Antenna Unit (IAU) and the stop/start switch must be pressed.

On each shut face of each door there is an emergency locking aperture. This is for use with the emergency access key in the event of a failure of the vehicle battery to lock the vehicle. Remove the cover and insert the key into the aperture to mechanically lock each door in turn, the driver's door being the last door to be locked. Please ensure that the key is removed from the emergency locking aperture before each door is closed.
For additional information, refer to: Handles, Locks, Latches and Entry Systems (501-14 Handles, Locks, Latches and Entry Systems, Description and Operation).

DESCRIPTION

DOOR MODULES

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The door modules provide the interface between the door latches and the Body Control Module/Gateway (BCM/GWM) assembly. The door modules provide door latches status information and enable the door motors on request from the BCM/GWM assembly.

The rear door modules are also controlled by the BCM/GWM assembly. Additionally, the front door modules also control the exterior mirror functions.

BODY CONTROL MODULE/GATEWAY MODULE ASSEMBLY

E189833

The Body Control Module/Gateway Module (BCM/GWM) controls the following functions:

- The horns
- The tailgate latch
- The tailgate ajar switch
- The turn signal indicators
- The fuel filler flap operation.

The BCM/GWM also has a connection to the Restraints Control Module (RCM) for automatic operation of the interior lights and the turn signal indicators in the event of an accident.

NOTE:

> If the BCM/GWM assembly is replaced, the new module will require configuring to the master car configuration using a Land Rover approved diagnostic system.

The BCM/GWM assembly automatically arms and disarms the active anti-theft system when the vehicle is locked and unlocked after successful confirmation that a valid smart key has been used.

INSTRUMENT CLUSTER

The Body Control Module/Gateway Module (BCM/GWM) assembly controls the alarm indicator which is incorporated in the main display in the Instrument Cluster (IC).

The IC also controls the engine immobilization, in conjunction with the BCM/GWM assembly, the Powertrain Control Module (PCM) and the Anti-Lock Brake System (ABS) control module. The PCM controls the engine crank and fuel functions and the ABS control module controls the tilt function. The PCM and ABS control module communicate to each other after the BCM/GWM processes the valid smart key information.

Alarm indication

The alarm warning indicator is a red Light Emitting Diode (LED) located in the Instrument Cluster (IC). When the ignition is switched off (power mode 0), the indicator gives a visual indication of the active anti-theft system to show if the alarm system is set or unset. When the ignition is switched on, the indicator provides a visual indication of the status of the passive anti-theft (engine immobilization) system. If the immobilization system is operating correctly, the LED will be illuminated for 3 seconds at ignition on and then extinguish.

If a fault exists in the immobilization system, the LED will be either permanently illuminated or flashing for 60 seconds. This indicates that a fault exists and a Diagnostic Trouble Code (DTC) has been recorded. After the 60 second period, the LED will flash at different frequencies which indicate the nature of the fault.

Operation of the alarm indicator is controlled by the IC which varies the flash rate of the LED to indicate the system status of the alarm and the immobilization systems.

Operation of the alarm indicator is controlled by the BCM/GWM assembly.

ALARM/IMMOBILIZATION STATUS	ALARM INDICATOR STATUS	ALARM INDICATOR FUNCTION

UNSET	No flash	LED remains off
SET - Perimeter alarm mode	Flashing	LED will flash once every 2 seconds
SET - Perimeter with alarm sensing mode	Flashing	LED will flash once every 2 seconds
ACTIVE - Triggered	Flashing	LED will flash once every 2 seconds
UNSET - alarm activated during previous SET cycle	No flash	LED remains off

PASSIVE SOUNDER

E189836

The passive sounder is connected directly to the Body Control Module/Gateway Module (BCM/GWM) assembly. The BCM/GWM activates the passive sounder as a secondary security sounder when the alarm is triggered

BATTERY BACK-UP SOUNDER

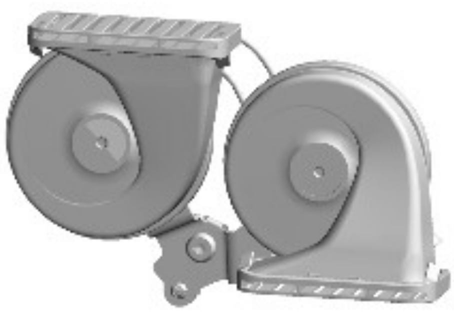
E189831

The Battery Back-Up Sounder (BBUS) is only fitted in certain markets and is mandatory fit in the UK.

The BBUS has an integrated tilt sensor which monitors the vehicle altitude. The Body Control Module/Gateway Module (BCM/GWM) assembly monitors the tilt sensor and can detect if the vehicle is being moved, towed or raised and will communicate to the BCM/GWM assembly in order to indicate a 'trigger' to activate the alarm.

Operation (SET/UNSET) of the BBUS and tilt sensor is controlled by the Body Control Module/Gateway Module (BCM/GWM) assembly on the Local Interconnect Network (LIN) bus. The BBUS is also connected with a permanent battery supply via the BCM/GWM assembly. An integral, rechargeable battery powers the sounder if the vehicle battery supply from the BCM/GWM assembly is interrupted.

VEHICLE HORNS



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The vehicle horns are located above the front bumper armature. The horns have a switched power supply via the horn relay located in the Engine Junction Box - 1 (EJB). The horns are switched via the steering wheel, connected through the Clockspring (CLKSPG) to the Body Control Module/Gateway Module (BCM/GWM). The BCM/GWM activates the horns as a secondary security sounder when the alarm is triggered.

INTERIOR MOTION SENSOR

E189838

The interior motion sensor (volumetric sensing) is located in a central position in the front overhead console.

Double Locking Vehicles

The volumetric sensors are activated when the vehicle is double locked. The vehicle can be locked and alarmed with the volumetric sensors deactivated if a pet is to be left in the vehicle either by:

- Single locking the anti-theft active system.
- Deactivating the alarm sensors in the instrument cluster settings menu.

Single locking

–In certain markets (Benelux) the volumetric sensors (where fitted) are activated on single locking as per latest market requirements.

When volumetric sensors are active and the vehicle battery voltage falls below 9 Volts, the BCM/GWM ignore any inputs from the sensors to prevent false alarm activation

The BCM/GWM ignores the signals from the volumetric sensor for the first 15 - 30 seconds to allow time for the vehicle interior to settle and prevent false alarm activation.

If the tailgate is opened via the smart key, the volumetric sensor and the tilt sensor are inhibited until the tailgate is closed.

HOOD SWITCH

The hood switch is attached to the underside of the right hood latch and operated by movement of the latch mechanism. When the latch opens, the switch closes and connects a ground to the Body Control Module/Gateway Module (BCM/GWM).

OPERATION

The Body Control Module/Gateway Module (BCM/GWM) assembly automatically arms and disarms the anti-theft system when it operates the Central Door Lock (CDL) system. The BCM/GWM also locks and unlocks the Electric Steering Column Lock (ESCL) (if equipped) when it operates the CDL system

On vehicles without a volumetric sensor, only the perimeter mode is available to monitor the hinged panels and the validity of the smart key.

When perimeter sensing is active, the BCM/GWM monitors aperture ajar switches located in the latch mechanisms of the front and rear doors, the tailgate and hood ajar micro switch, located in hood latch mechanism in the engine compartment.

When volumetric sensors are active, the BCM/GWM monitors the interior of the vehicle for movement using a volumetric sensor located the front overhead console

When the tilt sensors are active, the BCM/GWM monitors the vehicle for raising using an inclination sensor located within the Battery Back-up Sounder (BBUS).

ARMING

Perimeter mode

On vehicles without a volumetric sensor, or inclination sensor the anti-theft active system is armed in the perimeter mode when the vehicle is:

- Either locked or double locked using the lock switch on the smart key.
- On vehicles with the passive entry system, the lock/unlock switch on one of the exterior door handles.

Smart key switch selection and the lock/unlock switch selection on the exterior door handles are relayed to the Body Control Module/Gateway Module (BCM/GWM). The signal is sent by the Remote function Actuator (RFA) on the High Speed (HS) Controller Area Network (CAN) body systems bus.

Perimeter sensing only monitors the hinged panels and validity of the smart key in the Remote Function Actuator (RFA).

Perimeter with alarm sensing mode

Volumetric sensing monitors the vehicle interior for intrusion. If the vehicle is fitted with a Battery Back-up Sounder (BBUS), which incorporates a tilt sensor, the vehicle altitude is also monitored when perimeter with alarm sensing mode is active. Perimeter with alarm sensing mode is activated by a second press of the lock switch on the smart key or, on vehicles with the passive entry system, the lock switch on one of the exterior door handles. The second press of the switch must occur within 3 seconds of the first press. The second press of the lock switch also activates the perimeter sensing double locking feature.

In certain markets (Benelux) the volumetric sensors (where fitted) are activated on single lock as per latest market requirements.

The BCM/GWM arms the active anti-theft system when it single locks or double locks the vehicle, providing all the following conditions are met:

- All doors, tailgate and hood are closed
- The BCM/GWM is not in transit mode.

When the vehicle has successfully completed its locking routine, confirmation will be given by a single short flash of the turn signal indicators to indicate a single locked condition. If double locking is activated, then the confirmation will be given by a double flash of the turn signal indicators, one short flash for locked and one long flash on completion of double locked. If 'audible lock warning' is enabled, an audible chirp shall also be emitted from Battery Back-up Sounder (BBUS) on double lock. This feature can be turned on and off in the instrument cluster security features menu.

Mis-lock Audible Warning

If any doors, tailgate or hood is open/ajar, or if in Power Mode 4 or above, when a lock or double lock request is received, the anti-theft system remains disarmed. The Body Control Module/Gateway Module (BCM/GWM) generates a short mis-lock sound from the Battery Back-Up Sounder (BBUS) or passive sounder (turn signal indicators will not flash). Each attempt to lock will be confirmed by an audible chime being emitted.

If the tailgate is opened via the smart key, the volumetric sensor and the tilt sensor in the BBUS are inhibited until the tailgate compartment lid is closed.

Disarming Anti-theft System

The Body Control Module/Gateway Module (BCM/GWM) will disarm the anti-theft system to prevent false alarm activation under certain conditions as follows:

- When the anti-theft system is armed in anti-theft with an active system fitted with volumetric sensors:
- If the vehicle battery voltage decreases to less than 9 volts, the BCM/GWM will disable the perimeter with alarm sensing mode.
- The alarm will remain in perimeter mode only.

This prevents false alarm activation because the volumetric sensor cannot operate correctly below 9 volts.

On vehicles fitted with a Battery Back-Up Sounder (BBUS), if the vehicle battery voltage decreases from 9.5 to 9 volts in more than a 30 minute period, the BCM/GWM de-activates the BBUS. If required, the vehicle horns will sound an audible alarm trigger warning. This prevents false alarm activation.

At voltages below 9 volts, the BCM/GWM will not generate the 'heartbeat' signal to the BBUS. The BBUS interprets this as the BCM/GWM has been tampered with and activates the sounder. If the battery voltage subsequently rises to more than 9.5 volts, the BCM/GWM will re-arm the BBUS.

The BCM/GWM automatically re-locks the vehicle and rearms the active anti-theft system if:

- The vehicle is unlocked using the smart key
- And within 40 seconds a hinged panel is not opened

This prevents leaving the vehicle unlocked and disarmed by accidental operation of the smart key unlock switch.

Alarm

When the alarm is triggered, the BCM/GWM assembly activates audible and visual warnings. The audible warnings are produced by the Battery Back-Up Sounder (BBUS) or passive sounder (depending on market specification). Visible indication is provided by the turn signal indicators.

The BCM/GWM assembly activates the BBUS or passive sounder and vehicle horns (market dependant) and the visual indications for 30-60 seconds depending on the market.

The activation is stopped for 10 seconds and, if the alarm trigger is still present, the BCM/GWM assembly will cycle again for 30-60 seconds. This will be repeated for up to a maximum of 10 cycles per trigger source (3 cycles in certain markets) of 30 seconds (60 seconds in certain markets) for any one arming period.

The BCM/GWM assembly will ignore the trigger cause if the 10 cycles (3 cycles in certain markets) have been completed and the alarm trigger is still present or until it receives a disarm signal.

NOTE:

> If the Battery Back-Up Sounder (BBUS) is triggered due to tamper detection, the visual indication using the turn signal indicators is not activated.

The alarm can be triggered by any of the following:

- A hinged panel being opened.
- The doors unlocked.
- The volumetric sensor detects a movement inside the vehicle.
- The tilt sensor detects vehicle movement.
- An ignition tamper is detected (invalid smart key).

NOTE:

> If a BBUS is fitted, it is also armed with the anti-theft - active volumetric and tilt sensors mode lock request. However, the tilt functionality is not enabled in the anti-theft - active volumetric and tilt sensor mode.

On receipt of the arming signals, the BBUS and the tilt sensor respond with a status message.

If the BCM/GWM assembly does not receive the status signals within a period of time, the BCM/GWM assembly assumes there is a fault and responds with a disarm signal to either the BBUS or the tilt sensor and stores a related Diagnostic Trouble Code (DTC). If the BBUS is disarmed when the anti-theft system is armed and the system is subsequently triggered, the BCM/GWM assembly still energizes the horn relay and uses the horn to sound the audible warning in place of the BBUS.

When the sounder is armed, the BCM/GWM assembly sends a periodic (heartbeat) signal to the BBUS which prompts the BBUS to monitor the vehicle battery supply and the Local Interconnect Network (LIN) bus link with the BCM/GWM assembly. The BBUS will operate if:

- It receives an alarm signal from the BCM/GWM assembly or the tilt sensor.
- The power supply or the LIN bus link to the BCM/GWM assembly is disrupted.

The tilt sensor measures the longitudinal and lateral angle of the vehicle over a range of 16° degrees from the horizontal. When the anti-theft system is armed in perimeter with alarm sensing mode, the tilt sensor stores the current vehicle angles in its memory and monitors the tilt sensor readings. If the vehicle angle changes in either direction by more than the alarm limit threshold, the tilt sensor communicates to the BCM/GWM assembly in order to activates the BBUS.

If the alarm system is active and the battery or the BBUS is disconnected, the BBUS will continue to sound without the visual indication of the turn signal indicators flashing.

Panic Alarm

A panic alarm feature allows the vehicle alarm system to be activated using the smart key. The panic alarm switch, identified by a triangle symbol, can be operate and held for more than 3 seconds to activate the vehicle alarm.

To cancel the panic alarm feature perform one of the following:

- Operate the panic alarm switch three times.
- Press and hold the panic switch for 3 seconds.

- Press the stop/start switch.

To prevent accidental cancellation, the panic alarm cannot be cancelled within 5 seconds of being activated.

Smart key Additional Features

In addition to the lock and unlock switches, the smart key has convenience switches.

Headlamp Convenience

A headlamp convenience switch can be pressed to operate the headlamps to assist departure or approach to the vehicle. A single press of the switch will operate the headlamps for approximately 25 seconds, after which time they will automatically turn off. A second press of the switch will turn off the headlamps if the 25 second period has not been reached. Pressing the stop/start switch within the 25 second period will also turn off the headlamp convenience feature.

Convenience mode

When the vehicle is unlocked using the unlock switch on the smart key, the vehicle's electrical system initiates convenience mode.

The following systems become active in convenience mode:

- Memory - seat adjustment and mirror position
- Interior and exterior lighting
- Audio system
- IC message center
- Horn
- Cigar lighter/12V accessory socket.

Single Point Entry

The single point entry feature only unlocks the driver's door, all other doors remain single locked. A single press of the unlock switch on the Smartkey will unlock only the driver's door, a second press is required to unlock the remaining doors and tailgate.

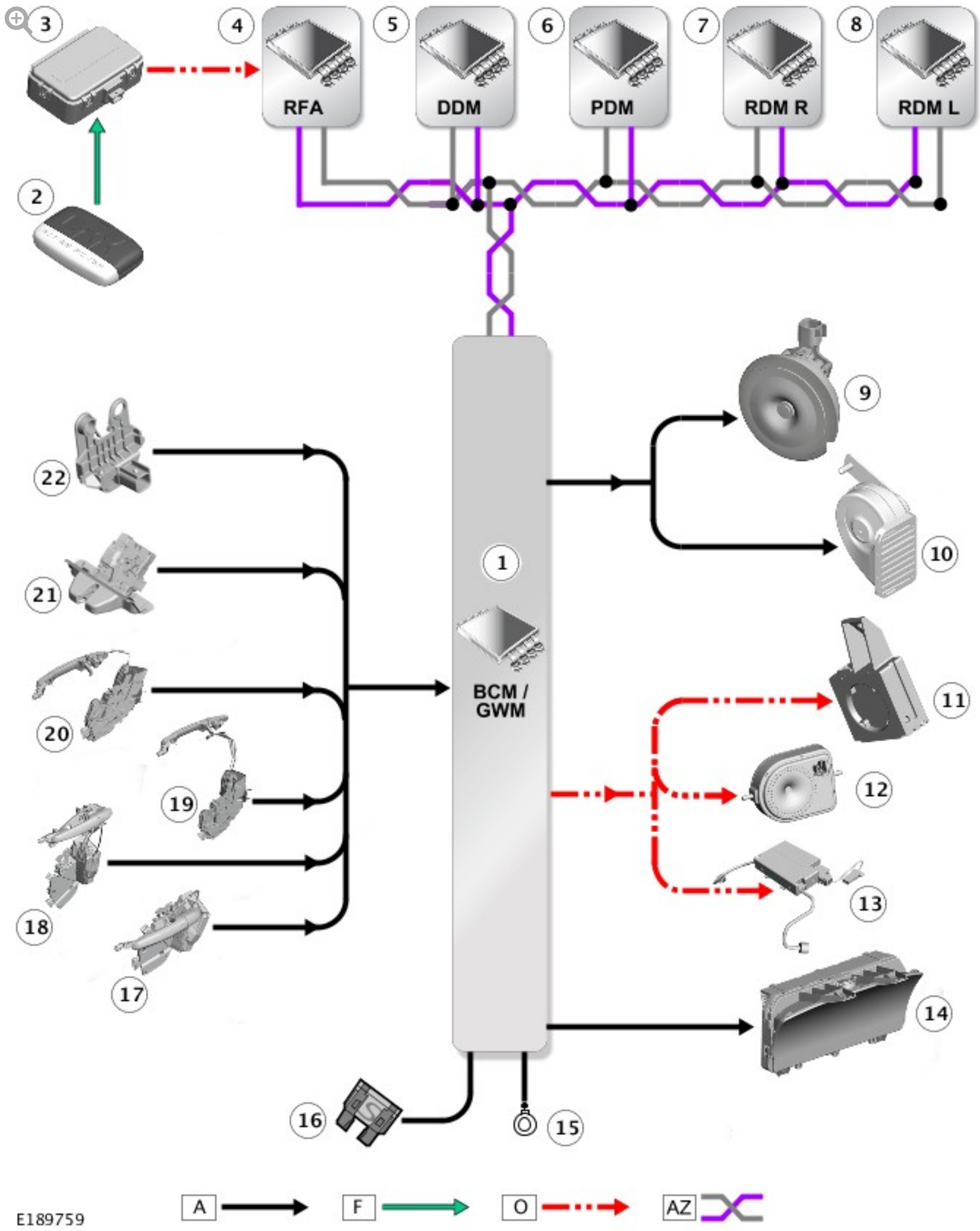
If the vehicle is double locked, the first press of the unlock switch on the Smartkey unlocks the driver's door. The remaining doors revert to the single locked state and can therefore be unlocked using the interior door handles, the Smartkey unlock switch or the unlock switch on the instrument panel switch pack.

Changing from central locking to single point entry can be carried out by pressing the lock and unlock switches on the Smartkey simultaneously. The turn signal indicators will flash to confirm that the function change has been performed.

Windows Global Open/Close

A global open and close feature can be operated from the Smartkey. This feature allows the vehicle windows to be opened/closed by a single press of the lock or unlock switch. The switch must be pressed and held for more than 2 seconds to activate the global open/close feature. The global open/close feature is not available in all markets. The windows must be initialized for the global functionality to work.

CONTROL DIAGRAM



A = HARDWIRED: F = RADIO FREQUENCY (RF) TRANSMISSION: O = LOCAL INTERCONNECT NETWORK (LIN): AZ = HIGH SPEED (HS) CONTROLLER AREA NETWORK (CAN) BODY BUS.

ITEM	DESCRIPTION
1	Body Control Module/Gateway Module (BCM/GWM)
2	Smart Key
3	Radio Frequency (RF) receiver

4	Remote Function Actuator (RFA)
5	Driver Door Module (DDM)
6	Passenger Door Module (PDM)
7	Rear Door Module (RDM) - Right
8	Rear Door Module (RDM) - Left
9	Passive sounder (where fitted)
10	Horn
11	Immobilizer Antenna Unit (IAU)
12	Battery Back-Up Sounder (BBUS) (market dependant)
13	Interior motion sensor (in overhead console)
14	Instrument Cluster (IC)
15	Ground
16	Supply
17	Front door latch mechanism with ajar switch - Driver
18	Front door latch mechanism with ajar switch - Passenger
19	Rear door latch mechanism with ajar switch - Right
20	Rear door latch mechanism with ajar switch - Left
21	Tailgate latch
22	Hood switch